Mechanics of Human Movement

Course Progress 90.2% (11 of 13)

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Assessment

Week 4: Assignment 4

Due date on 2026-08-19, 23:59 IST

The due date for submitting this assignment has passed.Your last recorded submission was on 2026-08-19, 22:11 IST.

1. The scientific study of the measurements of the human body is known as __________ 1 Point

○ Physiology
○ Anatomy
● Anthropometry
○ Kinesiology

Yes, the answer is correct. Score: 1 Accepted answer is Anthropometry

2. Anthropometric data consists of 1 Point

☑ Body segmental masses
☑ Body segmental lengths
☑ Centre of mass locations of body segments
☑ Body segmental shapes

No, the answer is incorrect. Score: 0

Accepted answer is Body segmental masses, Body segmental lengths, Centre of mass locations of body segments

3. The head being tipped backwards on the atlas is an example of ______ class lever in action 1 Point

● First
○ Second
○ Third
○ Fourth

Yes, the answer is correct. Score: 1 Accepted answer is First

4. The number of thoracic vertebra in the spinal column is: 1 Point

● 12
○ 7
○ 5
○ 1

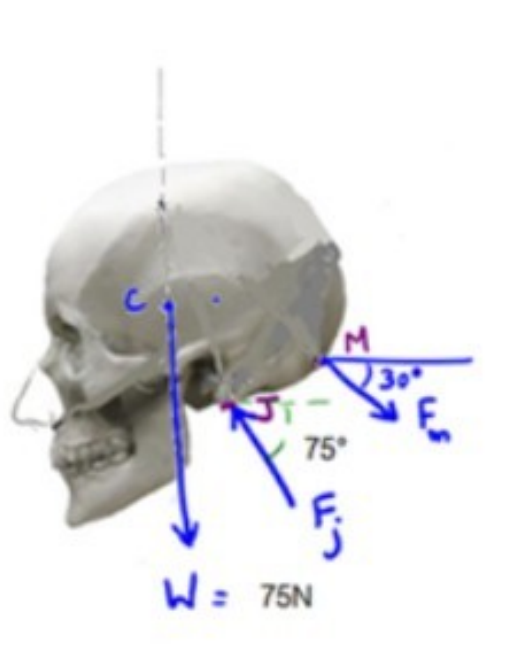
Yes, the answer is correct. Score: 1 Accepted answer is 12

5. While analysing the skull with the following planar force system: the joint force at the atlantooccipital joint, muscle force of the neck extensor and the weight of the skull, the forces form a _______ system of vectors 1 Point

☑ Concurrent
☐ Collinear
☐ Coplanar
☑ Non-concurrent

No, the answer is incorrect. Score: 0 Accepted answer is Concurrent, Coplanar

Use the adjacent diagram to answer questions 6-7



6. The tension in the neck extensor (F_m) for the configuration shown is ________ 1 Point

○ 43.1N
● 27.4 N
○ 35.3 N
○ 12.8 N

Yes, the answer is correct. Score: 1 Accepted answer is 27.4 N

7. The joint force at the atlantooccipital joint for this configuration is 1 Point

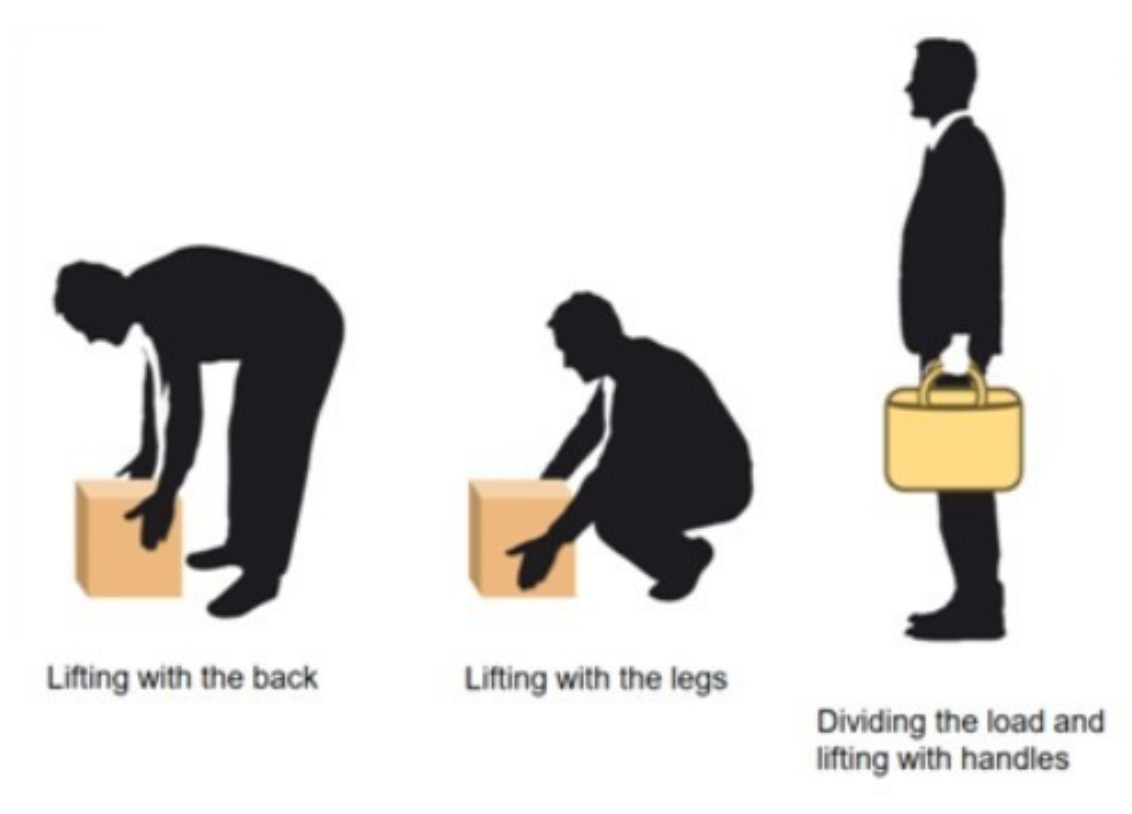
● 91.8 N
○ 52 N
○ 154 N
○ 35.3 N

Yes, the answer is correct. Score: 1 Accepted answer is 91.8 N

Use this information for questions 8-11.

A person (of body weight $W_B$and height H_B) is lifting a load W_L = 0.3 W_B in three different ways as shown in the figure below. To determine the best way of lifting, you have to do a static analysis of the spine to study the loads on it. Assume the following:

i. Length of trunk = 0.3 H_B
ii. Mass of trunk = 0.36 W_B
iii. Mass of head and arms = 0.18 W_B
iv. Centre of mass of trunk from distal end = 0.15 H_B
v. Point of action of muscle force from the distal end = 0.2 H_B
vi. The erector spinae is the only muscle acting on the spine at an orientation at 13° to the spine



8. If the thoracolumbar spine is inclined at an angle θ = 35˚for the first case, the force exerted by erector spinae will be 1 Point

○ 3.6 W_B
○ 2.09 W_B
● 4.5 W_B
○ 1.4 W_B

No, the answer is incorrect. Score: 0 Accepted answer is 3.6 W_B

9. The person is lifting a load of 0.3 W_B with the legs, as in the second case, such that the thoracolumbar spine is now inclined at an angle $\theta = 75°$. The force exerted by erector spinae will now be 1 Point

○ 0.77 W_B
○ 1.14 W_B
● 3.21 W_B
○ 4.56 W_B

No, the answer is incorrect. Score: 0 Accepted answer is 1.14 W_B

10. The person is now lifting a load of 0.3 W_B , which has been divided into two loads of 0.15 W_B each. In this case, the thoracolumbar spine is inclined at an angle, $\theta = 85°$. The force exerted by erector spinae will be 1 Point

○ 1.44 W_B
○ 4.56 W_B
● 0.38 W_B
○ 2.09 W_B

Yes, the answer is correct. Score: 1 Accepted answer is 0.38 W_B

11. Which of the following conditions is/are the safest for the spine while lifting the load? 1 Point

☐ Lifting with a bent back
☑ Lifting with an erect back
☐ Lifting without handles
☑ Dividing the load and lifting with both hands

Yes, the answer is correct. Score: 1 Accepted answer is Lifting with an erect back, Dividing the load and lifting with both hands

12. Which of the following motions can be performed by the shoulder joint? 1 Point

☑ Abduction
☐ Eversion
☑ Internal rotation
☑ Adduction

Yes, the answer is correct. Score: 1 Accepted answer is Abduction, Internal rotation, Adduction